

Finite and infinite series sheet

Gallo Tennis, CoGi / to A Mathematical Room

April 25, 2025

*Problems on series and products over finite and infinite sets: books of Terence Tao's "Analysis I", Donald E. Knuth's "The Art of Computer Programming".
As well as series inequalities problems from A. Engel's "Problem-Solving Strategies" and Andreescu's "Problems From The Book".*

Convergence of series

1. Let $\sum_{n=m}^{\infty} a_n$ be a formal series of real numbers. Then $\sum_{n=m}^{\infty} a_n$ converges if and only if, for every real number $\varepsilon > 0$, there exists an integer $N \geq m$ such that

$$\left| \sum_{n=p}^q a_n \right| \leq \varepsilon \quad \text{for all } p, q \geq N.$$

Solution.

- ($\implies$) If $\sum_{n=m}^{\infty} a_n$ converges then, by definition, the partial sum sequence

$$(S_N)_{N=m}^{\infty} = \left(\sum_{n=m}^N a_n \right)_{N=m}^{\infty}$$

converges to L . Therefore, for any $\varepsilon > 0$ there exists some $M \geq m$ such that for any $p-1, q \geq M$

$$|S_q - S_{p-1}| \leq \varepsilon.$$

Without loss of generality assume $p-1 \leq q$, it follows that

$$\left| \sum_{n=m}^q a_n - \sum_{n=m}^{p-1} a_n \right| = \left| \sum_{n=p}^q a_n \right| \leq \varepsilon.$$

- ($\impliedby$) Doing the reverse process of the direct implication the result follows.

□

2. **(Zero test).** Let $\sum_{n=m}^{\infty} a_n$ be a convergent series of real numbers. Then we must have $\lim_{n \rightarrow \infty} a_n = 0$. To put this another way, if $\lim_{n \rightarrow \infty} a_n$ is non-zero or divergent, then the series $\sum_{n=m}^{\infty} a_n$ is divergent.
- Solution.** If $\sum_{n=m}^{\infty} a_n$ converges then by last problem, for any $\varepsilon > 0$, there exists some $M \geq m$ such that for any $p, q \geq M$

$$\left| \sum_{n=p}^q a_n \right| \leq \varepsilon. \quad (1)$$

Suppose for the sake of contradiction that $\lim_{n \rightarrow \infty} a_n = L > 0$ so that for any $\varepsilon > 0$ there exists some M' such that for any $n \geq M'$

$$|a_n - L| < \varepsilon$$

so that in particular $a_n \geq L - \varepsilon$.

Therefore, for any $p, q \geq K := \max\{M, M'\}$

$$\left| \sum_{n=p}^q a_n \right| \geq \left| \sum_{n=p}^q L - \varepsilon \right| = (q - p + 1) |L - \varepsilon|.$$

Hence, let $\varepsilon = L/2$, we see that if the above inequality were true

$$\left| \sum_{n=p}^q a_n \right| \geq \frac{(q - p + 1)L}{2}.$$

Then let $q = p + 1$, it follows that

$$\left| \sum_{n=p}^q a_n \right| \geq L > \varepsilon.$$

A contradiction of inequality (1).

□

3. **(Absolute convergence test).** Let $\sum_{n=m}^{\infty} a_n$ be a formal series of real numbers. If this series is absolutely convergent, then it is also conditionally convergent. Furthermore, in this case we have the triangle inequality

$$\left| \sum_{n=m}^{\infty} a_n \right| \leq \sum_{n=m}^{\infty} |a_n|.$$

Solution. If $\sum_{n=m}^{\infty} |a_n|$ converges to L , then for any $\varepsilon > 0$ there exists some $M \geq m$ such that for any $p, q \geq M$

$$\left| \sum_{n=p}^q |a_n| \right| \leq \varepsilon.$$

Hence,

$$\sum_{n=p}^q |a_n| \leq \varepsilon.$$

By triangle inequality over finite series, for any $p, q \geq M$ we have

$$\left| \sum_{n=p}^q a_n \right| \leq \sum_{n=p}^q |a_n|.$$

We conclude that $\sum_{n=p}^q a_n$ converges. Trivially, we also note that again, by triangle inequality over finite series, each partial sum $S_N = \sum_{n=m}^N a_n$ is smaller than or equal to $T_N = \sum_{n=m}^N |a_n|$, meaning that by comparison principle $\lim_{N \rightarrow \infty} S_N \leq \lim_{N \rightarrow \infty} T_N$.

□

4. **(Series laws).**

- (a) If $\sum_{n=m}^{\infty} a_n$ is a series of real numbers converging to x , and $\sum_{n=m}^{\infty} b_n$ is a series of real numbers converging to y , then $\sum_{n=m}^{\infty} (a_n + b_n)$ is also a convergent series, and converges to $x + y$.

In particular, we have

$$\sum_{n=m}^{\infty} (a_n + b_n) = \sum_{n=m}^{\infty} a_n + \sum_{n=m}^{\infty} b_n.$$

- (b) If $\sum_{n=m}^{\infty} a_n$ is a series of real numbers converging to x , and c is a real number, then $\sum_{n=m}^{\infty} (ca_n)$ is also a convergent series, and converges to cx .

In particular, we have

$$\sum_{n=m}^{\infty} (ca_n) = c \sum_{n=m}^{\infty} a_n.$$

- (c) Let $\sum_{n=m}^{\infty} a_n$ be a series of real numbers, and let $k \geq 0$ be an integer. If one of the two series $\sum_{n=m}^{\infty} a_n$ and $\sum_{n=m+k}^{\infty} a_n$ are convergent, then the other one is also, and we have the identity

$$\sum_{n=m}^{\infty} a_n = \sum_{n=m}^{m+k-1} a_n + \sum_{n=m+k}^{\infty} a_n.$$

- (d) Let $\sum_{n=m}^{\infty} a_n$ be a series of real numbers converging to x , and let k be an integer. Then $\sum_{n=m+k}^{\infty} a_n$ also converges to x .

Solution. Let $T_N = \sum_{n=m}^N a_n$ and $R_N = \sum_{n=m}^N b_n$.

- (a) The partial sum of the series $\sum_{n=m}^{\infty} a_n + b_n$ is trivially

$$\sum_{n=m}^N (a_n + b_n) = T_N + R_N.$$

we also note that by limit laws

$$\lim_{n \rightarrow \infty} T_N + R_N = \lim_{n \rightarrow \infty} T_N + \lim_{n \rightarrow \infty} R_N.$$

We conclude that

$$\sum_{n=m}^{\infty} (a_n + b_n) = \lim_{N \rightarrow \infty} S_N = \lim_{N \rightarrow \infty} T_N + R_N = x + y.$$

- (b) Similarly to the item before, due to the linear properties of both limit and partial sums we get

$$\sum_{n=m}^N ca_n = cT_N$$

so that $\lim_{N \rightarrow \infty} S_N = \lim_{N \rightarrow \infty} cT_N = c \lim_{N \rightarrow \infty} T_N = cx$.

- (c) If $\sum_{n=m}^{\infty} a_n$ converges then the first claim is trivial, follows from convergent limit properties.

5. **(Telescoping series).** Let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers which converge to 0, i.e., $\lim_{n \rightarrow \infty} a_n = 0$. Then the series

$$\sum_{n=0}^{\infty} (a_n - a_{n+1})$$

converges to a_0 .

Solution. We see that for any partial sum S_N starting from 0,

$$\sum_{n=0}^N a_n - a_{n+1} = \sum_{n=0}^N a_n - \sum_{n=0}^N a_{n+1}$$

doing a substitution $n \rightarrow n + 1$ we get

$$S_N = \sum_{n=0}^N a_n - \sum_{n=1}^{N+1} a_n = a_0 + \sum_{n=1}^N a_n - \sum_{n=1}^N a_n + a_{N+1}$$

therefore,

$$S_N = a_0 - a_{N+1}.$$

We notice since (a_n) converges and by using limit laws that

$$\lim_{N \rightarrow \infty} S_N = \lim_{N \rightarrow \infty} a_0 - \lim_{N \rightarrow \infty} a_{N+1} = a_0.$$

□

6. (**Alternating series test.**) Let $(a_n)_{n=m}^{\infty}$ be a sequence of real numbers which are non-negative and decreasing, thus $a_n \geq 0$ and $a_n \geq a_{n+1}$ for every $n \geq m$. Then the series $\sum_{n=m}^{\infty} (-1)^n a_n$ is convergent if and only if the sequence a_n converges to 0 as $n \rightarrow \infty$.

Proof.

- ($\Rightarrow$) If $\sum_{n=m}^{\infty} (-1)^n a_n$ is convergent, then the sequence $((-1)^n a_n)_{n=m}^{\infty}$ converges to zero. By zero test for sequences, we know that the sequence $(|(-1)^n a_n|)_{n=m}^{\infty}$ converges to zero, and thus $(a_n)_{n=m}^{\infty}$ converges to zero.
- ($\Leftarrow$) Let any partial sum be $S_N = \sum_{n=m}^N (-1)^n a_n$. Let N be an even number. We claim that for any $k \geq 0$

$$S_{N+2k} \leq S_N.$$

To prove this claim we induct on k .

Basis. If $k = 0$ then

$$S_{N+2(0)} = S_N \leq S_N.$$

Hypothesis. Assume that for any positive integer k it follows that

$$S_{N+2k} \leq S_N.$$

Thesis. Our goal is to prove that for the integer $k + 1$ it follows that

$$S_{N+2(k+1)} \leq S_N.$$

The following chain of equalities will hold, by the recursive definition of a partial sum

$$S_{N+2k+2} = S_{N+2k} + (-1)^{N+2k+1} (a_{N+2k+1}) + (-1)^{N+2k+2} (a_{N+2k+2}).$$

Thus since N is even

$$S_{N+2k+2} = S_{N+2k} + (-1)(a_{N+2k+1}) + a_{N+2k+2}$$

note that $a_{N+2k+2} - a_{N+2k+1} \leq 0$ and thus

$$S_{N+2k+2} \leq S_{N+2k}.$$

By hypothesis and by transitivity we conclude that $S_{N+2k+2} \leq S_N$.

Seeing S_{N+2k} as a sequence when varying k , the previous result allows us to use it as a lower bound for S_N for any $N \geq m$. Consider now the dual odd sequence S_{N+2k+1} , for any $k \geq 0$ we observe that

$$S_{N+2k+1} = S_{N+2k} - a_{N+2k+1}$$

thus $S_{N+2k+1} \leq S_{N+2k}$. Furthermore we claim that

$$S_{N+2k+1} \geq S_{N+1}.$$

To prove this claim, by induction over k

Basis. If $k = 0$ then

$$S_{N+1} \geq S_{N+1}$$

so this case holds.

Hypothesis. Assume that for any $k \geq 0$ it holds

$$S_{N+2k+1} \geq S_{N+1}.$$

Thesis. Our goal is to prove that for the positive integer $k + 1$

$$S_{N+2(k+1)+1} \geq S_{N+1}$$

holds. To verify this

$$S_{N+2k+3} = S_{N+2k+1} + (a_{N+2k+2} - a_{N+2k+3}) \geq S_{N+2k+1}.$$

By inductive hypothesis and by transitivity we get

$$S_{N+2k+3} \geq S_{N+2k+1} \geq S_{N+1}.$$

We can arrange our sequences into

$$S_{N+2k} \geq S_{N+2k+1} \geq S_{N+1}$$

and therefore

$$S_N + a_{N+1} \geq S_{N+2k+1} \geq S_N - a_{N+1}.$$

So

$$|S_{N+2k+1} - S_N| \leq a_{N+1}$$

which implies that

$$\left| \sum_{n=N+1}^{2k+1} (-1)^n a_n \right| \leq a_{N+1}.$$

Since a_{N+1} converges to zero and is decreasing, for any $\varepsilon > 0$ we can find some $M \geq m$ such that for any $n \geq M$ $a_n \leq \varepsilon$. By problem (1), since $N + 1$ and $2k + 1$ are arbitrary, this series converges.

□

7. **(Comparison test.)** Let $\sum_{n=m}^{\infty} a_n$ and $\sum_{n=m}^{\infty} b_n$ be two formal series of real numbers, and suppose that $|a_n| \leq b_n$ for all $n \geq m$. Then if $\sum_{n=m}^{\infty} b_n$ is convergent, then $\sum_{n=m}^{\infty} a_n$ is absolutely convergent, and in fact

$$\left| \sum_{n=m}^{\infty} a_n \right| \leq \sum_{n=m}^{\infty} |a_n| \leq \sum_{n=m}^{\infty} b_n.$$

Proof. Since (b_n) is convergent, the partial sum sequence

$$(S_N)_{N=m}^{\infty} = \left(\sum_{n=m}^N b_n \right)_{N=m}^{\infty}$$

is bounded by a positive real number K since it converges and $0 \leq |a_n| \leq b_n$.

The partial sum sequence $(T_N)_{N=m}^{\infty} = \left(\sum_{n=m}^N |a_n| \right)_{N=m}^{\infty}$ is a positive terms sum and therefore it is increasing. Notice that each of the partial sums will be less than or equal to the partial sums S_N (over the corresponding indices).

Since it is increasing and it is bounded above by K it converges, and it will converge to a smaller limit to that of S_N .

□

8. **(Geometric series.)**

9. **(Cauchy criterion.)** Let $(a_n)_{n=1}^{\infty}$ be a decreasing sequence of non-negative real numbers (so $a_n \geq 0$ and $a_{n+1} \leq a_n$ for all $n \geq 1$). Then the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if the series

$$\sum_{k=0}^{\infty} 2^k a_{2^k} = a_1 + 2a_2 + 4a_4 + 8a_8 + \cdots$$

is convergent.

Solution.

- ($\Rightarrow$) If $\sum_{n=1}^{\infty} a_n$ converges then (a_n) converges to zero “from above”, this means that for any $\varepsilon > 0$ we can find some $n_{\varepsilon} \geq m$ such that $a_n \leq \varepsilon$. Hence, since (a_n) is decreasing

$$\sum_{n=0}^N a_n \leq a_0 \varepsilon.$$

In fact, we also know that since $\sum_{n=1}^{\infty} a_n$ is a positive terms series, there exists some $M \geq 0$ such that for any $N \geq 0$

$$\sum_{n=0}^N a_n \leq M.$$

($\Leftarrow$)

10. Let $\sum_{n=0}^{\infty} a_n$ be an absolutely convergent series of real numbers such that $\sum_{n=0}^{\infty} |a_n| = 0$. Show that $a_n = 0$ for every natural number n .
11. Let $\sum_{n=0}^{\infty} a_n$ be an absolutely convergent series of real numbers. Let $f: \mathbb{N} \rightarrow \mathbb{N}$ be an increasing function (i.e., $f(n+1) > f(n)$ for all $n \in \mathbb{N}$). Show that $\sum_{n=0}^{\infty} a_{f(n)}$ is also an absolutely convergent series. (Hint: try to compare each partial sum of $\sum_{n=0}^{\infty} a_{f(n)}$ with a (slightly different) partial sum of $\sum_{n=0}^{\infty} a_n$.)
12. **(Root test.)**
13. **(Ratio test.)**
14. Show that a number $x \in \mathbb{R}$ is rational if and only if its q -ary expression in any base q is periodic, that is, from some rank on it consists of periodically repeating digits.

Solution.

($\Rightarrow$) If $x \in \mathbb{R}$ is a rational number, then

($\Leftarrow$) Let $q \geq 1$. Let $\mathbb{N}_q$ be the q -ary representation of the natural numbers set. If x has a periodic q -ary expression, then there exists some $a \in \mathbb{N}_q$ such that

$$x = p + \sum_{n=m}^{\infty} a \left(\frac{1}{q^n} \right) = p + a \sum_{n=m}^{\infty} \left(\frac{1}{q^n} \right).$$

in other words, from m all the decimal places become a , note that p represents the integer part of x .

To test out our definition we need to show that for $m, N \in \mathbb{N}_q$ it holds

$$x = m/N = p + a \sum_{n=m}^{\infty} \left(\frac{1}{q^n} \right).$$

Let $m = pN + Q$, remains showing that $Q/N = a \sum_{n=m}^{\infty} \left(\frac{1}{q^n} \right)$. Since $q \geq 1$, the previous series is geometric, and thus

$$a \sum_{n=m}^{\infty} \frac{1}{q^n} = a \left(\frac{1}{1 - \frac{1}{q}} \right) = a \left(\frac{q}{q-1} \right).$$

This means that we can choose $N = \frac{q-1}{a}$.

Series classification

1. Let $(a_n)_{n \geq 1}$ be a sequence of positive real numbers such that the series $\sum_{n=1}^{\infty} a_n$ converges. Show that the series

$$\sum_{n=1}^{\infty} a_n^{\frac{n}{n+1}}$$

Also converges.

2. Show that

$$(a) \quad S_0(n) = 1^0 + \cdots + n^0 = n,$$

$$S_1(n) = 1^1 + \cdots + n^1 = \frac{n(n+1)}{2} = \frac{1}{2}n^2 + \frac{1}{2}n,$$

$$S_2(n) = 1^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6} = \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6}n,$$

$$S_3(n) = \frac{n^2(n+1)^2}{4} = \frac{1}{4}n^4 + \frac{1}{2}n^3 + \frac{1}{4}n^2,$$

and in general that

$$S_k(n) = a_{k+1}n^{k+1} + \cdots + a_1n + a_0$$

is a polynomial in n of degree $k+1$.

$$(b) \quad \lim_{n \rightarrow \infty} \frac{S_k(n)}{n^{k+1}} = \frac{1}{k+1}.$$

Solution.

(a) By induction over k we claim

3. Show that

(a) the equality

$$1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!} + \frac{1}{n!n} = 3 - \frac{1}{1 \cdot 2 \cdot 2!} - \cdots - \frac{1}{(n-1) \cdot n \cdot n!}$$

holds for $n \geq 2$;

$$(b) \quad e = 3 - \sum_{n=0}^{\infty} \frac{1}{(n+1)(n+2)(n+2)!};$$

(c) for computing the number e approximately the formula

$$e \approx 1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!} + \frac{1}{n!n}$$

is much better than the original formula

$$e \approx 1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!}.$$

Power series and generating functions

Application

1. König's theorem.